SciOdyssey.

Autonomous ML Research Agent

Preserve the research world. Reset the researcher.

01 / METHOD

Different lifetimes. Shared evidence.

Long-lived reasoning can anchor the search; rebuilding context is costly. SciOdyssey retains the world while reopening judgment with a fresh Scientist.

Research environment
Task contract + research preferences
Data + evaluator + baseline · budget + permissions

↓

Research world

PERSISTENT / ON DISK

EVIDENCE	**Knowledge + experiment ledger** Metrics, failures and original reports
CURATED	**Research brief + fallible priors** Current understanding; open to revision
CODE	**Retained implementation + State** Recoverable version with evidence and provenance

↓ CONTEXT RESET Inherit experience, not prior reasoning.

Scientist → META

Scientist
SCIENTIFIC JUDGMENT

- Read the research world
- Form hypotheses
- Code, train and evaluate
- Interpret + report evidence

Chooses the science within one trajectory.

META
PERSISTENCE AUTHORITY

- Audit claims ↔ evidence
- Scope conclusions
- Curate shared memory
- Preserve optional State

Maintains what survives across trajectories.

↻ Audit → scope → preserve → fresh Scientist

Runtime / deterministic mechanics
Isolation · cancellation · runner + evaluator
Logging · artifact capture · State operations

One world. Several independent searches.

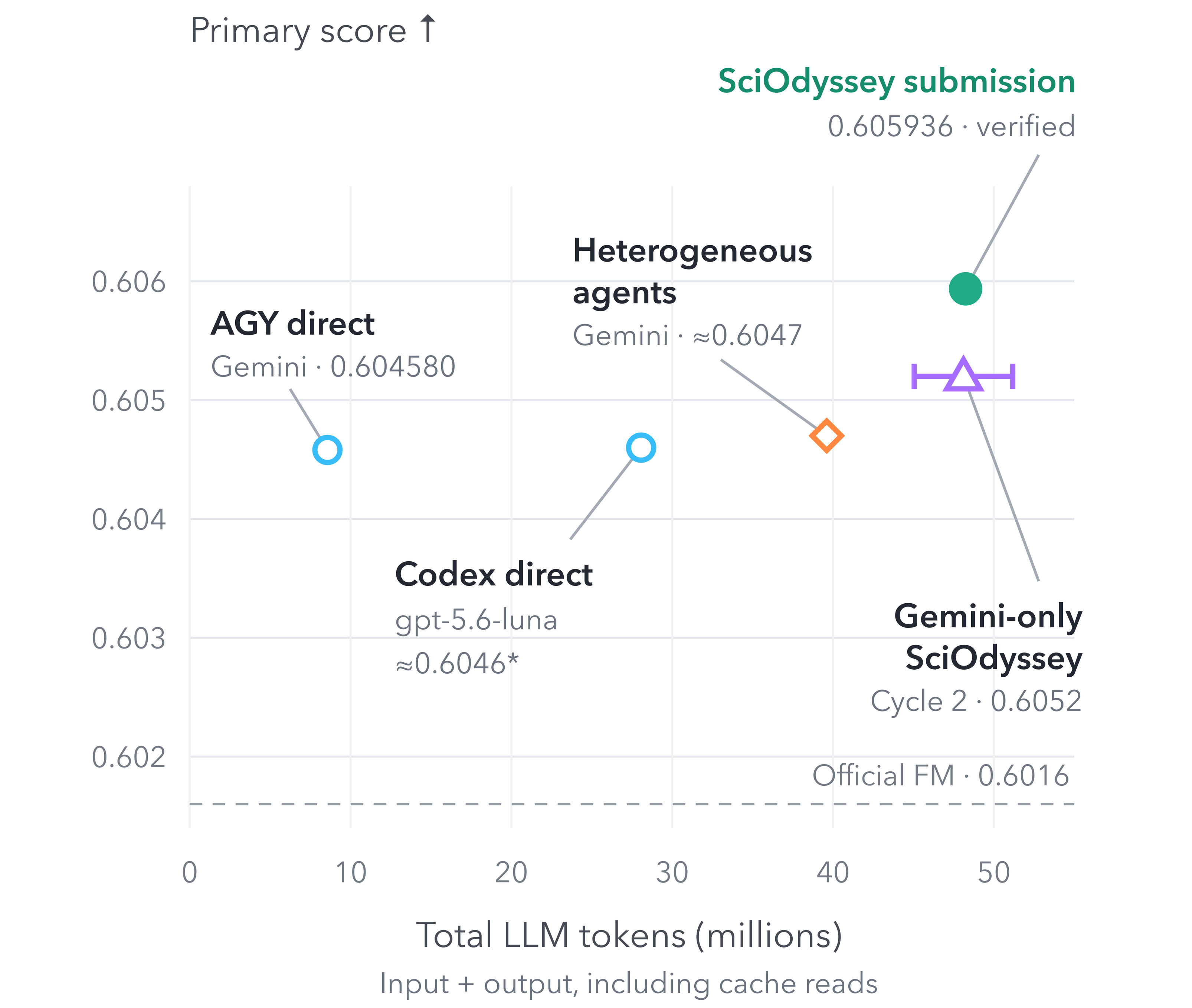
Isolated worktrees, then post-hoc review. Optional synthesis keeps one implementation parent and reference evidence; code and memory are not auto-merged.

Framework capability, not an isolated cause of benchmark gain.

02 / EVIDENCE

Research outcomes, in context.

KuaiRand-Pure · public validation · one recorded run per condition



* Codex is provisional. Gemini = gemini-3.7-flash. The Cycle-2 interval is a token-accounting bound. Models and protocols differ: no causal, significance or token-efficiency claim is made.

Verified submission · public validation

Metric	Official FM	SciOdyssey	Absolute Δ
Primary	**0.6016000**	**0.6059363**	**+0.0043363**
GAUC	0.6674000	0.6728421	+0.0054421
nDCG@5	0.5357000	0.5390304	+0.0033304

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical features × eight-seed FM ensemble · NumPy / CPU.

4 autonomous cycles · 13 named experiments · 0 manual scientific interventions*

7 Full evaluations · 0 GPU-hours · 170,588 prediction rows passed the unchanged Starter Kit alignment checker.

* After launch, in the retained submission record.

Observed behavior, not just a score.

Recovery
After evaluation, NumPy float32 values broke serialization. The Scientist repaired the writer and retained the evidence.

Audit
Reviews found leakage and sampling mismatches. META scopes evidence; it is not a formal verifier.

Open questions. One task does not establish generality. META, State and reset effects are not isolated; summaries can still lose context and anchor future work.

03 / YOUR RESEARCH WORKFLOW

One agent. Any ML task. End to end.

Autonomous by default. Supervisable by design.

`task.md`
What to solve
Objectives · constraints · evaluation

`PERSONAL.md`
How you want it solved
Preferences · research style · priorities

Define the task → Delegate research → Review evidence

Coding agent — EXAMPLE INPUT

```
› Use the research-agent skill for task.md.
  Follow PERSONAL.md. Run autonomously.
  Let me review the results and evidence.
```

github.com/zc6600/research-agent-kuairand ↗

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Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md §§3-5 · Code, report, ledger and checked output in repository.